

From “God” to Infrastructure: The Self-Expanding Logic of Artificial Intelligence, Energy, and Space Industry

A Technological-Civilizational Framework Based on the “Colonizer-Indigenous” Analogy

Abstract

This paper explores a structural analogy between historical colonial encounters and the emerging relationship between artificial intelligence (AI) and human civilization. The analogy is not intended as a historical or moral equivalence. It is a framework for examining how technological power can move from mystery to demystification, reproduction, infrastructure, and expansion. The central argument is that the transformative potential of AI may depend less on whether AI simply exceeds humans on cognitive benchmarks than on whether machine intelligence can be integrated with energy production, computation, robotics, manufacturing, resource extraction, communications, and space infrastructure. This creates a possible feedback loop: energy enables computation; computation enables AI; AI improves automation and manufacturing; automation expands infrastructure and resource access; and the resulting infrastructure enables additional energy and computation. The paper calls this process industrial self-expansion or technological bootstrapping rather than literal self-replication.

1. From “God” to “Human”: The Demystification of Technology

When radically unfamiliar technology first appears, people often understand it through existing cultural categories. A capability that cannot yet be explained may seem supernatural or “godlike.” Once its mechanisms become visible, its limitations, resource requirements, and operating conditions become clearer. AI is undergoing a similar process: users increasingly recognize hallucinations, context limits, dependence on data and computation, and the importance of tools and infrastructure.

2. What Determines Power: Intelligence or Reproducibility?

Technological power does not arise from intelligence alone. It arises when capabilities can be reproduced, coordinated, scaled, and connected to material resources. AI is unusual because cognitive capability itself is becoming increasingly reproducible: a trained model can be copied, deployed simultaneously, connected to tools, and integrated into workflows. Intelligence can therefore become infrastructure.

3. From Tool to Infrastructure

A tool performs a task; infrastructure creates the conditions under which many tasks can continuously occur. AI may become a distributed layer within software development, scientific research, industrial design, logistics, energy management, robotics, finance, education, and manufacturing. Its significance may therefore lie less in isolated chatbot interactions than in its integration into production systems.

4. Energy: The Physical Foundation of AI Expansion

Digital intelligence is not immaterial. AI depends on semiconductor fabrication, servers, data centers, electricity, cooling, networking, buildings, and supply chains. A useful conceptual chain is: Energy → Computation → Intelligence → Automation → Manufacturing → Infrastructure → More Energy and Computation. If AI improves energy forecasting, grid management, photovoltaic

deployment, battery manufacturing, and industrial automation, it may indirectly accelerate the growth of the physical systems that support AI itself.

5. Why Silicon Is an Interesting Bridge

Silicon is central to both photovoltaics and semiconductor electronics, although photovoltaic-grade and semiconductor-grade silicon involve different specifications and processes. This creates an interesting conceptual connection: Sunlight → photovoltaic materials → electricity → semiconductor computation → AI. The significance is not that silicon solar automatically powers AI, but that energy capture and computation can belong to the same broader technological ecosystem.

6. Space Infrastructure as a Second Expansion Dimension

Space infrastructure introduces another dimension of expansion. Reusable launch systems, satellite communications, lunar resource extraction, robotic construction, space-based solar power, and orbital data centers represent different layers of a possible future industrial stack. Their economic and engineering feasibility varies greatly, and many remain uncertain. The important threshold would be the emergence of an industrial ecosystem capable of maintaining and expanding infrastructure beyond Earth.

7. From Terrestrial Industry to Space Industry

A space industry would initially depend heavily on Earth. Industrial bootstrapping asks how much of the next generation of infrastructure could eventually be produced from resources already available in the new environment. If robotic systems could extract lunar materials, manufacture structural components, produce energy systems, and assemble new machines, each generation could reduce dependence on Earth. Perfect self-replication is not required; sufficiently high local production and resource recycling may be enough.

8. Why “Self-Replication” May Be the Wrong Term

Modern industry does not reproduce as a single organism. It consists of specialized factories, mines, power plants, software systems, logistics networks, and communication systems. A more realistic concept is networked self-expansion: one facility produces machines that enable other facilities, which produce energy systems and materials for additional facilities. Expansion occurs through specialization and interdependence rather than literal duplication.

9. What Changes When AI Enters Industrial Bootstrapping?

AI may reduce constraints in design, simulation, coordination, software development, scientific discovery, and industrial optimization. Connected to robotics, it can translate digital decisions into physical action. A possible feedback structure is: Energy → Compute → AI → Design and Coordination → Robotics and Manufacturing → Infrastructure → Resources and Energy → More Compute. The system does not need to be conscious to be self-expanding; what matters is whether the technological network can increase its capacity over time.

10. Material-Basis Expansion Capability

A useful conceptual variable is Material-Basis Expansion Capability (MBEC): $MBEC = f(E, C, R, M, S, A)$, where E is energy availability, C computing capacity, R robotics and automation, M manufacturing capability, S resource access, and A autonomous coordination. This is not a physical law but a framework for identifying conditions required for technological expansion. High computational intelligence with weak physical infrastructure remains constrained; strong industrial infrastructure can exert enormous systemic power even with modest intelligence.

11. The Expansion Ratio

Another conceptual measure is the Industrial Expansion Ratio: $R = \Delta K / K$, where K is productive infrastructure and ΔK is the increase in productive capacity over a given period. The purpose is to ask whether an industrial system can consistently generate more productive infrastructure than is required to establish it. The key question becomes whether AI increases the rate at which civilization can build the infrastructure required for further growth.

12. From AI Alignment to Infrastructure Alignment

AI safety discussions often focus on whether advanced systems' goals are aligned with human values. The infrastructure perspective adds another layer: who controls the physical expansion loop? Energy → Chips → Compute → AI → Robotics → Manufacturing → Resources → Energy. Control over these interfaces can become strategically significant. The long-term governance problem may therefore involve the distribution of control across an increasingly automated industrial network.

13. Humans Are Not the Indigenous Population of the Analogy

The colonizer-indigenous analogy has strict limits. Historical colonizers and indigenous societies were distinct human populations with independent histories. AI is a human-created technological system. Humans simultaneously occupy the roles of creators, users, owners, subjects affected by AI, and potential competitors with AI-enabled organizations. The analogy is therefore structural rather than historical or moral.

14. Who Controls the Expansion Loop?

The popular question “Will AI rule humanity?” may be too narrow. A more structurally useful question is: Who controls the expansion loop? If energy, computation, robotics, manufacturing, communications, and resource extraction become tightly integrated with AI, power may accumulate around the actors controlling the interfaces between these systems. The outcome is not predetermined: the same architecture could support centralized control, decentralized production, cooperative infrastructure, or combinations of these forms.

15. Conclusion

The deepest transformation brought by AI may not be the creation of a machine that simply thinks better than humans. It may be scalable machine cognition participating in the expansion of the physical infrastructure required for cognition itself. The historical metaphor of moving from “god” to “ordinary human” illustrates technological demystification, but demystification does not necessarily eliminate systemic power. The critical transition is: Mystery → Understanding → Reproduction → Infrastructure → Automation → Expansion. If AI becomes deeply coupled with solar energy, advanced manufacturing, robotics, and eventually space infrastructure, the result could be a technological system capable of extending its own material basis. Such a system would not need literal self-replication; it could be a networked industrial system in which intelligence, energy, manufacturing, and resource acquisition reinforce one another.

Selected Research Directions

- Carbon-neutral and space-based data centers
- Scalable space-based AI infrastructure
- Affordable and rapid bootstrapping of the space industry
- Robotic assembly and autonomous construction of large space structures

- Autonomous lunar industrialization and space-based solar power